

Year 10
Pathway 2



Baldivis
Secondary College

Mini Test - Simultaneous Equations

Name: Sol. Ootions
~~Solutions~~

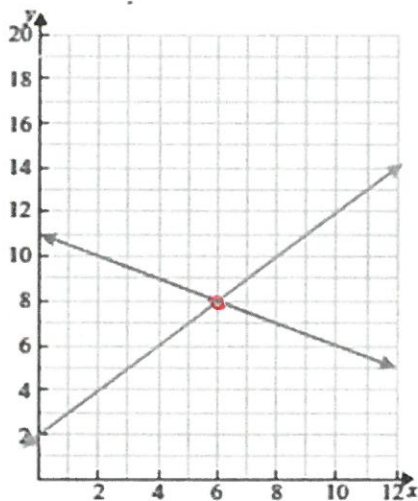
Total Marks: 30 marks
Total Time: 30 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Resources allowed: 1 page of notes A4, 1 side. Calculator

1. Determine the point of intersection for the following pairs of simultaneous equations.

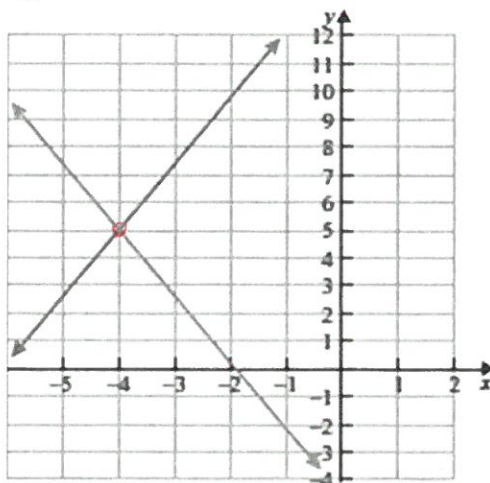
a.



(6, 8)

(1 mark)

b.

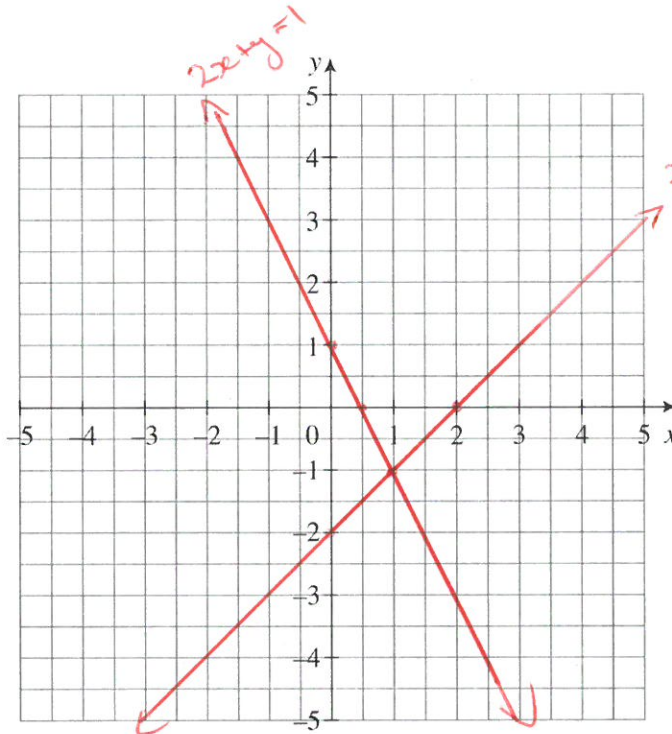


(-4, 5)

(1 mark)

2.

- Sketch the graphs with equations $x - y = 2$ and $2x + y = 1$ accurately on the set of axes provided.
- Solve the equations graphically by stating the x and y values at the point of intersection.
- Solve the equations simultaneously (algebraically) and verify your graphical answer.



$x - y = 2 \leftarrow a$
 $2x + y = 1$
 $(1, -1)$
 $x = 1 \quad y = -1$
 $\hline$
 $c) \begin{array}{r} x - y = 2 \\ + \quad 2x + y = 1 \\ \hline 3x + 0 = 3 \\ x = 1 \end{array}$
 $2(1) + y = 1$
 $2 + y = 1$
 $y = -1$
 $\therefore x = 1, y = -1$
 (5 marks)

3. Find the solution to the following pairs of simultaneous equations by substitution.

a. $3a + b = 8$
 $b = a - 12$

$3a + (a - 12) = 8$
 $4a = 20$
 $a = 5$

$b = 5 - 12$
 $b = -7$
 $\therefore a = 5 \text{ and } b = -7$

b. $c = 7 - 2d$
 $3c - 2d = 9$

$3(7 - 2d) - 2d = 9$
 $21 - 6d - 2d = 9$
 $21 - 8d = 9$
 $-8d = -12$
 $d = \frac{3}{2}$

$c = 7 - 2\left(\frac{3}{2}\right)$
 $= 4$

(6 marks)

$\therefore d = \frac{3}{2} \text{ and } c = 4$

4. Find the solution to the following pairs of simultaneous equations by elimination.

a. $x + y = 5$

$+ x - y = 1$

$2x = 6$

$x = 3$

$3 + y = 5$

$y = 2$

$\therefore x = 3 \text{ and } y = 2$

b. $5x + y = 12 \times 2$

$3x + 2y = 3$

$10x + 2y = 24$

$- 3x + 2y = 3$

$7x = 21$

$x = 3$

$3(3) + 2y = 3$

$9 + 2y = 3$

$2y = -6$

$y = -3$

$\therefore x = 3 \text{ and } y = -3$ (5 marks)

5. Solve each of the following pairs of simultaneous equations:

a. $3x + 4y = -5$

$x - y = 3 \times 3$

$3x + 4y = -5$

$- 3x - 3y = 9$

$7y = 14$

$y = 2$

$x - 2 = 3$

$x = 5$

$\therefore x = 5 \text{ and } y = 2$

b. $2x - 3y = 11$

$y = 2x - 1$

$2x - 3(2x - 1) = 11$

$2x - 6x + 3 = 11$

$-4x = 8$

$x = -2$

$y = 2(-2) - 1$
 $= -5$

$\therefore x = -2 \text{ and } y = -5$

(6 marks)

6. The length of a rectangle (y) is 6 cm longer than its width (x). The perimeter of the rectangle is 102 cm.
- a. Form two equations.

$$y = x + 6$$

$$2x + 2y = 102$$

- b. Solve the equations using substitution and so state the length and width of the rectangle.

$$2x + 2y = 102$$

$$y = x + 6$$

$$2x + 2(x + 6) = 102$$

$$2x + 2x + 12 = 102$$

$$4x = 90$$

$$x = 22.5$$

$$y = 22.5 + 6$$

$$= 28.5$$

(6 marks)

∴ length is 28.5 cm and
width is 22.5 cm